(a) Prefix reuse

Saved interaction

1. Build | 2. Append | 3. Edit early | 4. Restore

Prompt excerpt

```
You are maintaining a Python repository. Diagnose the timeout failure below.

deadline = monotonic() + timeout
connection = await client.acquire()
```

Reused / prompt tokens



Model response Restore

The root cause is that the `timeout` parameter is only used to set a deadline for the function's execution, but it is not enforced during the `client.acquire()` call.

512 reused tokens | TTFT **47.3 ms** | DRAM → GPU

KV Inspector

Independent export/import check
Block 1 · 128 tokens · 36 layers

BF16 **18.00 MiB** → Encoded blob **12.73 MiB**

0 BF16 mismatches

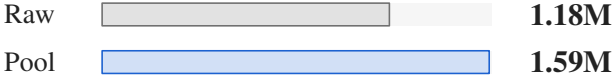
After Build · 576 records

(b) GPU capacity

Recorded

Admission: $S = 8$ | $S = 12$

KV tokens / GPU — 54 GiB budget



Waiting requests (mean): 1.5 → **0.4**

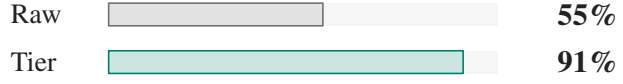
Engine TTFT (s): 5.16 → **1.94**

(c) History retention

Recorded

NVMe: 640 GiB | 2 TiB

Cached tokens (GPU + external)



Stored / raw bytes: **71.6%**

Engine TTFT (s): 6.76 → **2.34**